Assignment2

CORBA Viva Questions and Answers

1. What is CORBA?

CORBA stands for Common Object Request Broker Architecture.
It is middleware used for communication between distributed objects in different systems.

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2. What is the full form of ORB?

ORB stands for Object Request Broker.

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3. What is the role of ORB in CORBA?

ORB acts as a communication bridge between client and server objects.

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4. What is middleware?

Middleware is software that connects different applications or systems.

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5. What is distributed computing?

Distributed computing means multiple computers work together over a network.

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6. What is IDL in CORBA?

IDL stands for Interface Definition Language.
It defines methods and interfaces between client and server.

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7. What is the purpose of Calc.idl?

It defines calculator operations like addition, subtraction, multiplication, and division.

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8. What is static invocation?

In static invocation, the client already knows the server methods at compile time.

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9. What is dynamic invocation?

Dynamic invocation allows clients to discover interfaces during runtime.

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10. What are the advantages of CORBA?

- Reusability
- Portability
- Flexibility
- Distributed communication

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11. What is a servant in CORBA?

A servant is the server-side implementation object that performs requested operations.

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12. What is Naming Service in CORBA?

Naming Service helps clients locate remote objects using names.

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13. What is ORB.init()?

It initializes the ORB object.
Example:

ORB orb = ORB.init(args, null);

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14. What is narrow() in CORBA?

narrow() converts a generic CORBA object reference into a specific object type.

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15. What is POA in CORBA?

POA stands for Portable Object Adapter.
It manages server-side CORBA objects.

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16. What is the purpose of shutdown() method?

It stops the server application gracefully.

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17. Which operations are implemented in calculator application?

- Addition
- Subtraction
- Multiplication
- Division

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18. What does rebind() do?

It registers the server object with the Naming Service.

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19. What is the use of resolve_initial_references()?

It gets references to CORBA services like NameService and RootPOA.

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20. What is the purpose of client program?

The client sends requests to the server for calculations.

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21. What is the purpose of server program?

The server processes requests and returns results.

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22. What are the steps to execute CORBA application?

1. Create IDL file
2. Compile IDL
3. Create server
4. Create client
5. Compile programs
6. Start Naming Service
7. Run server
8. Run client

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23. Which package is used for CORBA in Java?

org.omg.CORBA

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24. What is the output of the calculator application?

It performs arithmetic operations and returns the result to the client.

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25. Difference between RMI and CORBA?

RMI	**CORBA**

RMI	CORBA
Java-specific	Language-independent
Simpler	More complex
Uses Java objects	Uses IDL
Platform dependent	Platform independent

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26. Why is CORBA platform independent?

Because it allows communication between applications written in different languages and running on different platforms.

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27. What is object brokering?

Object brokering means connecting client requests to remote server objects through ORB.

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28. What happens when orb.run() is called?

The server starts waiting for client requests.

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29. What is the aim of this practical?

To develop a distributed application using CORBA for object brokering.

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30. What is the final result of this practical?

Successfully created a CORBA system for distributed communication.